

EDA REPORT OF AUSTO MOTOR COMPANY

PROBLEM

Analysts are required to explore data and reflect on the insights. Clear writing skill is an integral part of a good report. Note that the explanations must be such that readers with minimum knowledge of analytics is able to grasp the insight.

Austo Motor Company is a leading car manufacturer specializing in SUV, Sedan, and Hatchback models. In its recent board meeting, concerns were raised by the members on the efficiency of the marketing campaign currently being used. The board decides to rope in an analytics professional to improve the existing campaign.

A. What is the important technical information about the dataset that a database administrator would be interested in? (Hint: Information about the size of the dataset and the nature of the variables)

Solution -A

Dataset: austo_automobile.csv

The dataset is about the Austo Motor Company is a leading car manufacturer specializing in SUV, Sedan and Hatchback models. Below are some important technical information about the dataset:

- The data set contains of 1581 rows & 14 columns.
- The data contains 5 integer, 1 float & 8 Object data type columns.

S. No.	Columns	Data Type
1	Age	int64
2	Gender	object
3	Profession	object
4	Marital_status	object
5	Education	object
6	No_of_Dependents	int64
7	Personal_loan	object
8	House_loan	object
9	Partner_working	object
10	Salary	int64
11	Partner_salary	float64
12	Total_salary	int64
13	Price	int64
14	Make	object

B. Take a critical look at the data and do a preliminary analysis of the variables. Do a quality check of the data so that the variables are consistent. Are there any discrepancies present in the data? If yes, perform preliminary treatment of data.

Solution -B

After a critical look at quality check the following discrepancies are noticed.

- Unique values in columns with object data type.

Columns	Data Type	Unique values
Gender	object	Male, Femal, Female, nan, Femle
Profession	object	Business, Salaried
Marital_status	object	Married, Single
Education	object	Post Graduate, Graduate
Personal_loan	object	No, Yes
House_loan	object	No, Yes
Partner_working	object	No, Yes
Make	object	SUV, Sedan, Hatchback

- It is noticed that in column **Gender** the spelling of **Female** is incorrectly mentioned adding 2 more unique values i.e. **Femal** & **Femle**.
- Below is the screenshot of the spelling error being fixed.

```
df.Gender = df.Gender.replace(["Femal", "Femle"], ["Female", "Female"])

# Replace mis spelled word with correct words

df.Gender.unique()

array(['Male', 'Female', nan], dtype=object)
```

- There are missing values in **Gender** & **Partner_salary** columns.

Columns	Data Type	No. Of Missing Values
Gender	object	53
Partner_salary	object	106

- Total No. of **Females** = **329** & **Males** = **1199** and there are **53 Nan**(null values).
- Since the number of **male customers** is **significantly higher than female customers**, the missing values in the **Gender** column were imputed with '**Male**'.
- The implementation of this step is shown in the figure below.

```
df.Gender = df.Gender.fillna("Male")

# Fill all the null value with mode which is Male

df.Gender.value_counts()

Gender
Male      1252
Female     329
Name: count, dtype: int64

df.Gender.isna().sum()

np.int64(0)
```

- The **Partner_salary** column contained **106 missing values**.
- Which were imputed by calculating the difference between **Total_salary** and **Salary**.
- The implementation of this step is shown in the figure below.

```
df.Partner_salary = df.Total_salary - df.Salary

# Fill Partner_salary as difference of Total_Salary and Salary

df.Partner_salary.isna().sum()

np.int64(0)
```

- The quantitative characteristics of the data set were examined, and no significant abnormalities or inconsistencies were observed in the numerical variables.

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
Age	1581	31.92	8.43	22	25	29	38	54
No_of_Dependents	1581	2.46	0.94	0	2	2	3	4
Salary	1581	60392.22	14674.83	30000	51900	59500	71800	99300
Partner_salary	1475	20225.56	19573.15	0	0	25600	38300	80500
Total_salary	1581	79626.00	25545.86	30000	60500	78000	95900	171000
Price	1581	35597.72	13633.64	18000	25000	31000	47000	70000

C. Explore all the features of the data separately by using appropriate visualizations and draw insights that can be utilized by the business.

Solution -C

Univariate Analysis Approach:

- Individual features were analyzed separately to understand their distribution and behavior.
- The data set contains **8 categorical variables** and **6 numerical variables**.
- Categorical variables represent customer demographics and purchase preferences and were analyzed using **bar charts**.
- Numerical variables represent age, income, and price-related information and were analyzed using **histograms and box plots**.
- The insights obtained from this analysis can be used to support business decision-making.

● **Feature-wise Analysis and Business Insights:**

➤ **Gender**

- **Visualization Used:** Bar chart
- **Observation:** The number of male customers is significantly higher than the number of female customers.
- **Business Insight:** Marketing strategies can focus primarily on male customers while designing targeted initiatives to increase female participation.

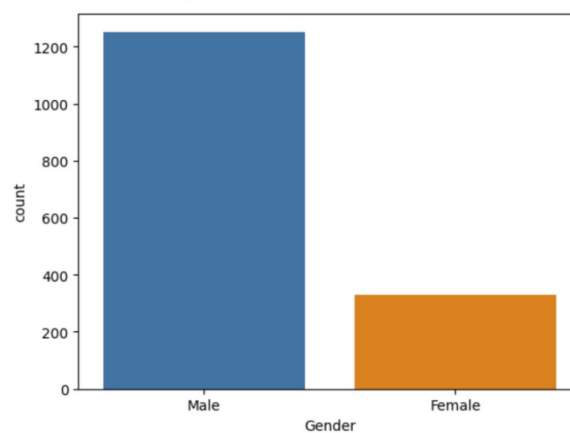


Figure: Distribution of customers by gender

➤ **Profession**

- **Visualization Used:** Bar chart
- **Observation:** Salaried professionals are slightly higher in number compared to business professionals; however, the difference between the two groups is relatively small.

- **Business Insight:** Since both salaried and business professionals contribute significantly to automobile purchases, marketing and sales strategies should be designed to address the needs of both segments rather than focusing exclusively on one group.

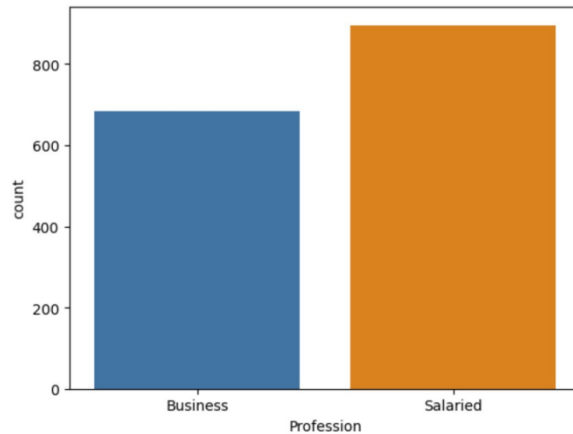


Figure: Distribution of customers by profession

➤ Marital Status

- **Visualization Used:** Bar chart
- **Observation:** The dataset contains a significantly higher number of **married customers** compared to **single customers**.
- **Business Insight:** Since married individuals form the majority of buyers, automobile companies can focus on promoting family-oriented vehicles and features, such as safety, comfort, and financing options tailored for households.

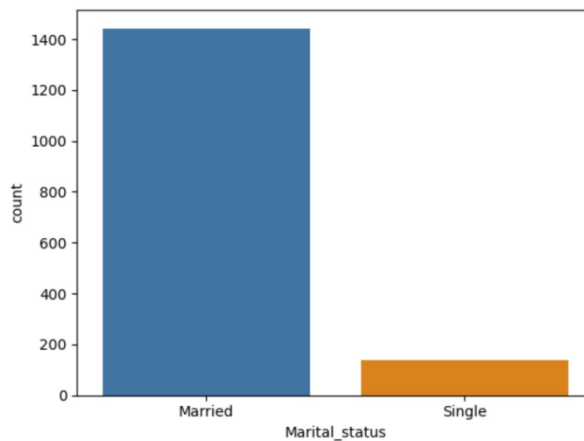


Figure: Distribution of customers by marital status

➤ Education

- **Visualization Used:** Bar chart
- **Observation:** The number of customers with **post-graduate education** is slightly higher than those with **graduate education**.
- **Business Insight:** A higher proportion of well-educated customers suggests that marketing communication can emphasize technical features, value propositions, and long-term benefits of automobiles.

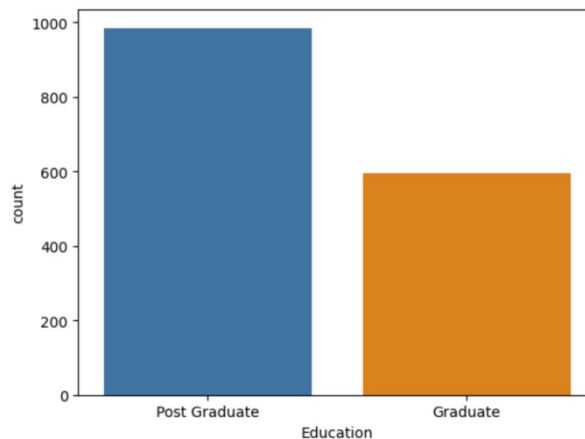


Figure: Distribution of customers by education

➤ Personal Loan

- **Visualization Used:** Bar chart
- **Observation:** The number of customers having a personal loan and those not having a personal loan is almost similar.
- **Business Insight:** Since both customers with and without personal loans actively purchase automobiles, marketing strategies and financing options should cater to both segments.

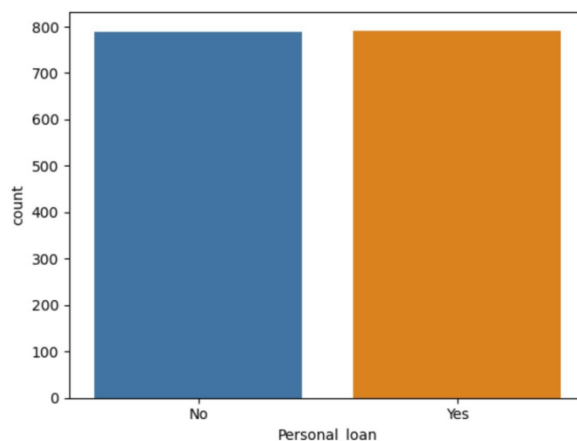


Figure: Distribution of customers by personal loan status

➤ House Loan

- **Visualization Used:** Bar chart
- **Observation:** The number of customers **without a house loan** is significantly higher than those having a house loan.
- **Business Insight:** Since customers without house loans form a larger segment, the company can focus on flexible pricing and financing options for this group.

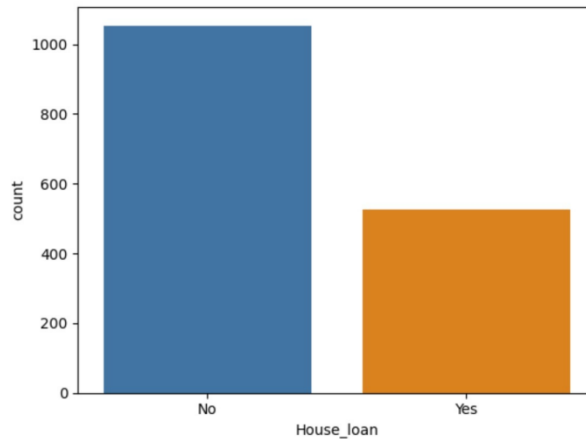


Figure: Distribution of customers by house loan status

➤ Partner Working

- **Visualization Used:** Bar chart
- **Observation:** Customers with a working partner are slightly higher in number compared to those whose partner is not working.
- **Business Insight:** Since both groups are well represented, **the company** should consider household income variations while designing pricing and financing strategies.

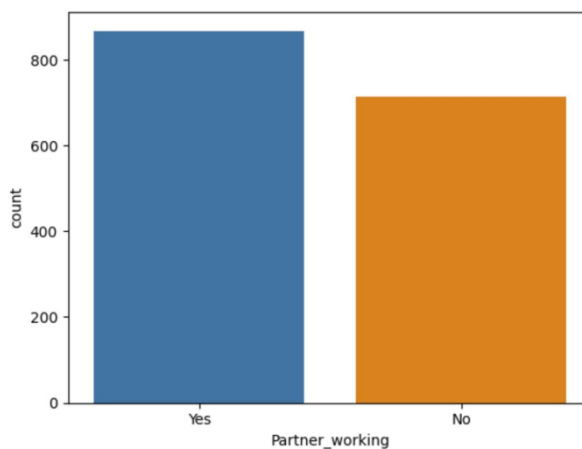


Figure: Distribution of customers based on partner working status

➤ Make

- **Visualization Used:** Bar chart
- **Observation:** **Sedan** is the most frequently purchased automobile type, followed by **Hatchback**, while **SUV** has the lowest number of purchases.
- **Business Insight:** Since sedans have the highest demand, **the company** should prioritize sedan models in production, inventory planning, and promotional activities.

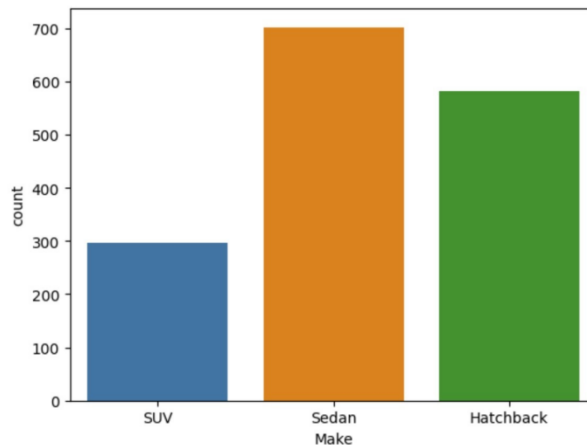


Figure: Distribution of automobile purchases by make

● **Analysis of Numerical Features and Business Insights:**

➤ Age

- **Visualization Used:** Histogram
- **Observation:** The age distribution indicates that most customers are between **22 and 30 years**, with the **median age around 29 years**. The number of customers gradually decreases as age increases, with fewer customers above 40 years.
- **Business Insight:** Since a majority of customers are young adults, **the company** can focus its marketing strategies, product features, and pricing models to suit the preferences of younger buyers.

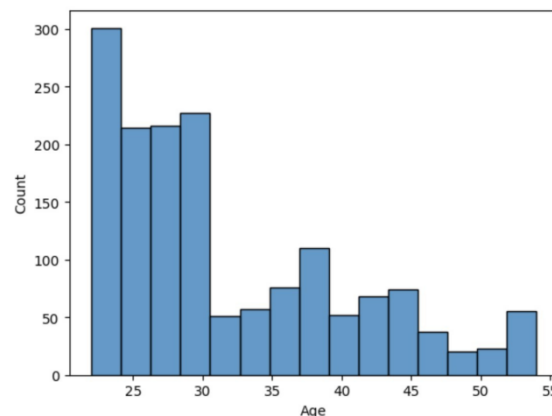


Figure: Age distribution of customers

➤ Number of Dependents

- **Visualization Used:** Histogram
- **Observation:** Most customers have **2 or 3 dependents**, with very few customers having **0 or 4 dependents**.
- **Business Insight:** Since the majority of customers have moderate family sizes, **the company** can focus on promoting vehicles that are suitable for small to medium-sized families, emphasizing space, comfort, and safety features.

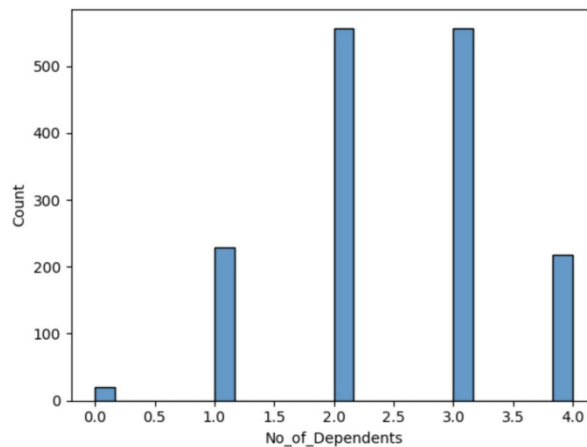


Figure: Distribution of customers by number of dependents

➤ Salary

- **Visualization Used:** Histogram
- **Observation:** The salary distribution shows that most customers earn between **₹50,000 and ₹65,000**, with a higher concentration around the mid-income range. A gradual decline is observed for higher salary values.
- **Business Insight:** Since the majority of customers fall within the mid-income group, **the company** can focus its pricing and financing strategies on affordability and value-for-money offerings.

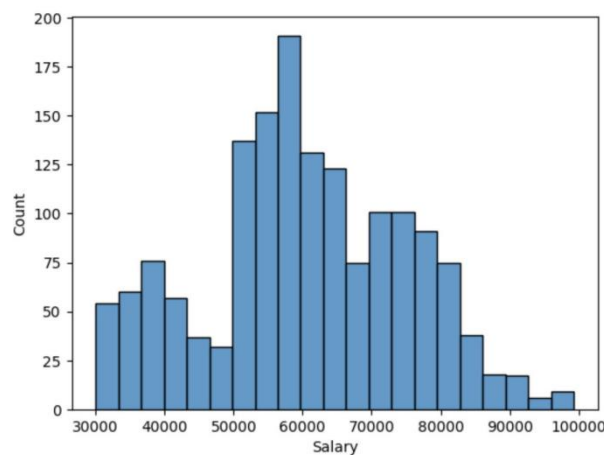


Figure: Distribution of customers by salary

➤ Partner Salary

- **Visualization Used:** Histogram
- **Observation:** The distribution of partner salary shows a large concentration at **zero**, indicating many customers with non-working partners. Among working partners, most salaries lie between **₹20,000 and ₹40,000**, with very few high-income cases.
- **Business Insight:** Since a significant portion of customers belong to single-income households, **the company** should consider affordable pricing and financing options, while also offering premium choices for dual-income households.

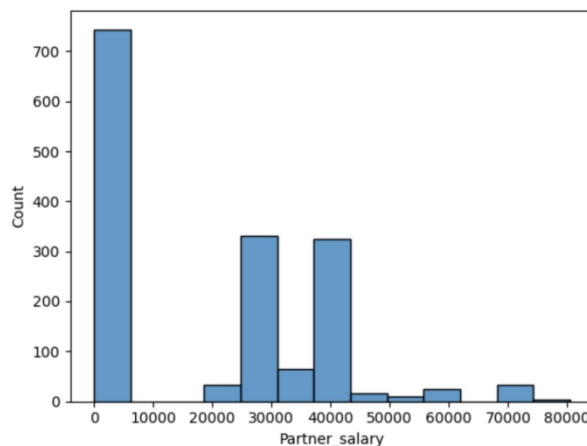


Figure: Distribution of partner salary

➤ Total Salary

- **Visualization Used:** Histogram
- **Observation:** Most customers have a total household income between **₹60,000 and ₹1,00,000**, with a median of around **₹78,000**. A small number of customers fall in the higher income range.
- **Business Insight:** Since most customers belong to the mid-income segment, **the company** can focus on mid-range vehicle offerings and flexible financing options, while selectively targeting higher-income households.

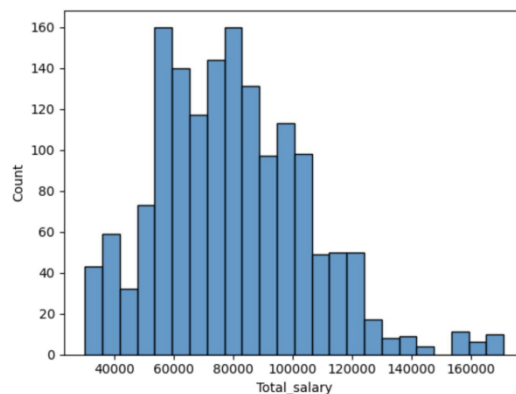


Figure: Distribution of customers by total salary

➤ Price

- **Visualization Used:** Histogram
- **Observation:** Most automobiles are priced between ₹20,000 and ₹30,000, with prices extending up to ₹70,000 and the number of purchases decreasing as price increases.
- **Business Insight:** Since the highest demand lies in the lower price range, **the company** should focus on competitively priced models while selectively promoting higher-priced vehicles.

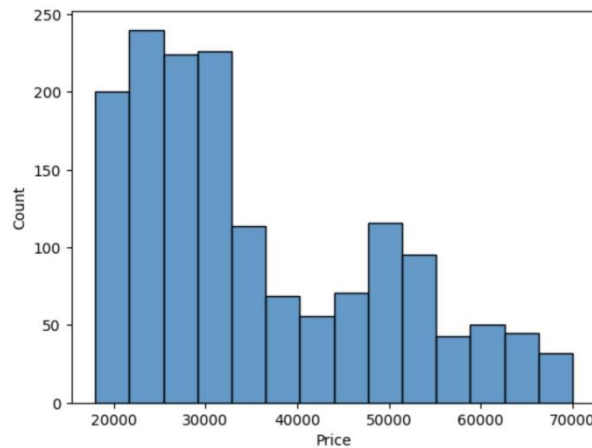


Figure: Distribution of automobile prices

D. Understanding the relationships among the variables in the dataset is crucial for every analytical project. Perform analysis on the data fields to gain deeper insights. Comment on your understanding of the data.

Solution -D

To understand the relationships among variables in the dataset, **bivariate and multivariate analysis** was performed. This analysis helps identify how customer demographics, income levels, and professional attributes influence automobile purchase behavior.

Bivariate Analysis Approach:

➤ Gender vs Automobile Make

Analysis Performed:

The dataset was balanced by randomly sampling **three subsets** with equal numbers of **male and female customers (329 each)**. The relationship between gender and automobile make was analyzed separately for **SUV, Hatchback, and Sedan**.

Observation

- **SUV:** Female customers show higher SUV purchase counts across all balanced samples.
- **Hatchback:** Male customers consistently show higher hatchback purchases across samples.
- **Sedan:** Sedan purchases show nearly equal preference between male and female customers.

Interpretation

The results indicate that **gender influences automobile preference differently across vehicle types**. Female customers prefer SUVs, male customers prefer hatchbacks, while sedans remain largely gender-neutral.

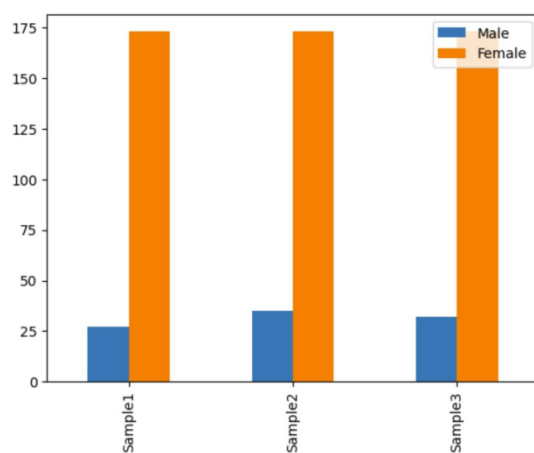


Figure: Gender-wise comparison of SUV purchases

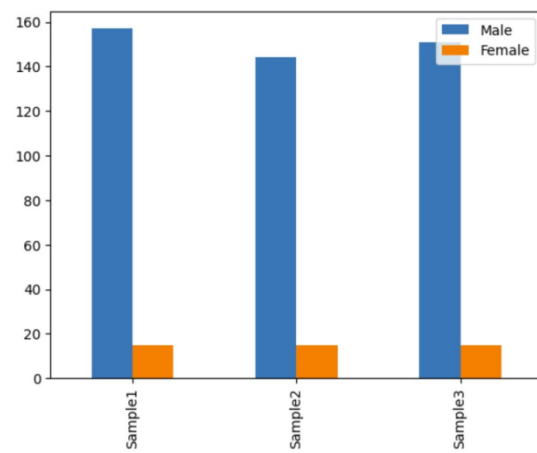


Figure: Gender-wise comparison of hatchback purchases

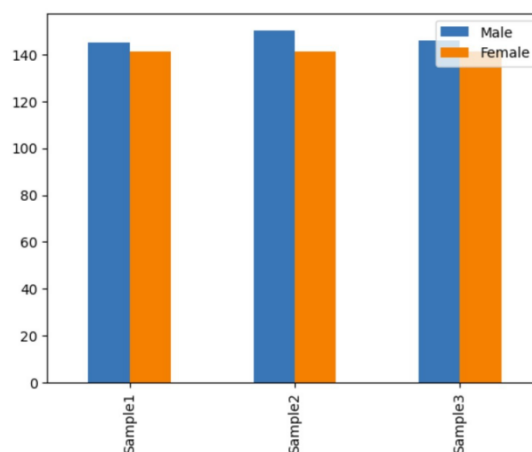


Figure: Gender-wise comparison of sedan purchases

➤ Profession vs Make

Analysis Performed

- The relationship between **Profession (Salaried and Business)** and **car Make (SUV, Hatchback, Sedan)** was analyzed.
- Since the number of salaried and business customers did not show a significant imbalance, **no sampling was required**.
- Bar charts were used to compare the preference of different car types across professions.

Observations

- **SUV:** Salaried customers show a higher preference for SUVs compared to business customers.
- **Hatchback:** Both salaried and business customers show almost equal interest in hatchback cars.
- **Sedan:** Sedan cars are preferred more by salaried customers than business customers.

Interpretation

- Overall, **salaried customers tend to purchase more vehicles across all car types**, especially sedans and SUVs.
- **Business customers show relatively balanced but lower purchase counts**, with a comparatively higher inclination toward hatchback cars.
- This indicates that profession influences car preference, with salaried customers generally contributing more to overall sales.

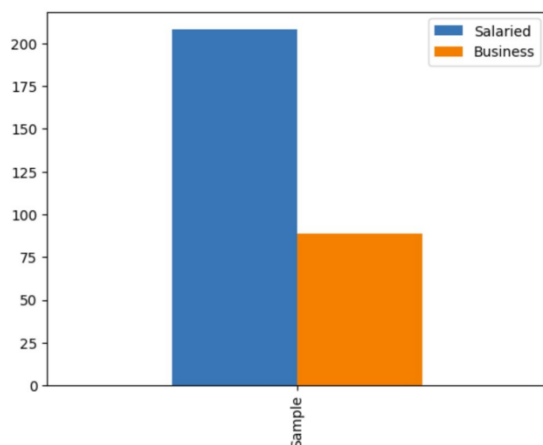


Figure: SUV purchases by profession

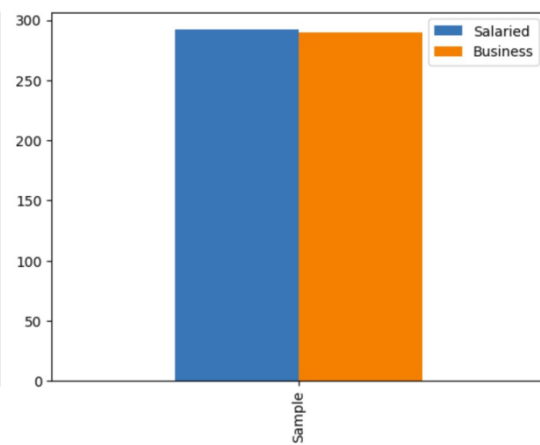


Figure: Hatchback purchases by profession

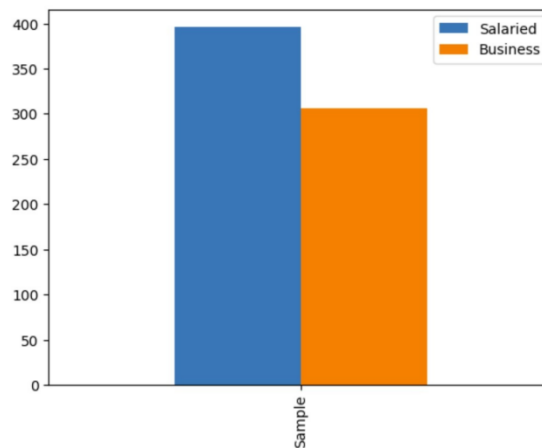


Figure: Sedan purchases by profession

➤ Marital Status vs Make

Analysis Performed:

The number of married customers was higher than single customers. To avoid bias, the data was balanced by creating five samples, each containing **138 married and 138 single customers**. Using these balanced samples, analysis was performed to study the relationship between marital status and vehicle type (SUV, Hatchback, and Sedan).

Observation:

- Married customers show a higher preference for **SUVs and Sedans** across all samples.
- Single customers show relatively higher preference for **Hatchbacks**.
- The trend remains consistent across all five samples, indicating stable patterns.

Interpretation:

Marital status influences vehicle preference. Married customers tend to prefer SUVs and Sedans due to family needs and comfort, while single customers prefer Hatchbacks because of affordability and convenience.

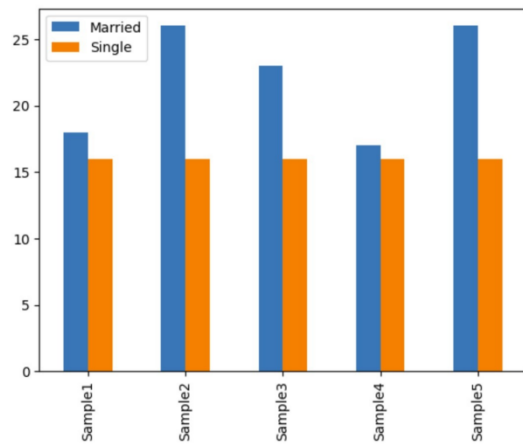


Figure: SUV preference by marital status

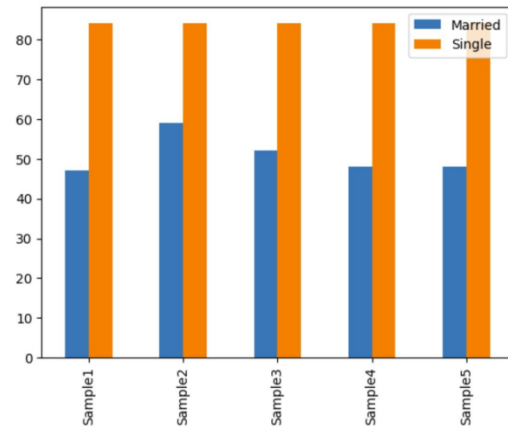


Figure: Hatchback preference by marital status

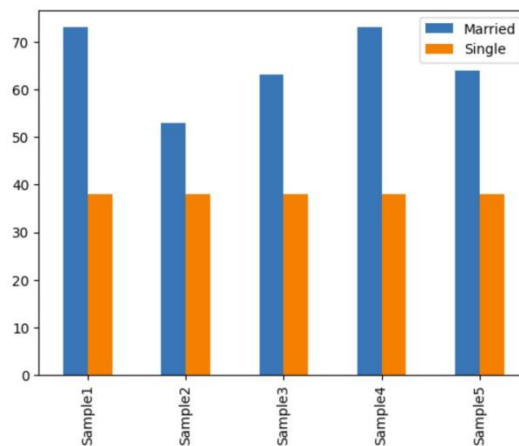


Figure: Sedan preference by marital status

➤ Personal Loan vs Make

Analysis Performed

- Customers **with and without personal loans** are almost equal in number.
- Hence, **no sampling was required**.
- Analysis was performed to understand the impact of **personal loan status** on the purchase of **SUV, Hatchback, and Sedan** cars.

Observation

- **SUV:**
Customers **without personal loans** slightly outnumber those with personal loans, but the difference is not large.
- **Hatchback:**
Purchases are **almost equally distributed** between customers with and without personal loans.

- **Sedan:**
A **slightly higher number of customers with personal loans** prefer sedans compared to those without loans.

Interpretation

- Personal loan status shows **limited influence** on overall car preference.
- Hatchbacks are purchased **independently of loan status**, indicating affordability.
- Sedans show a **mild inclination toward customers using personal loans**, suggesting financing support for higher-priced vehicles.
- Overall, **vehicle type preference is driven more by need and affordability than loan availability**.

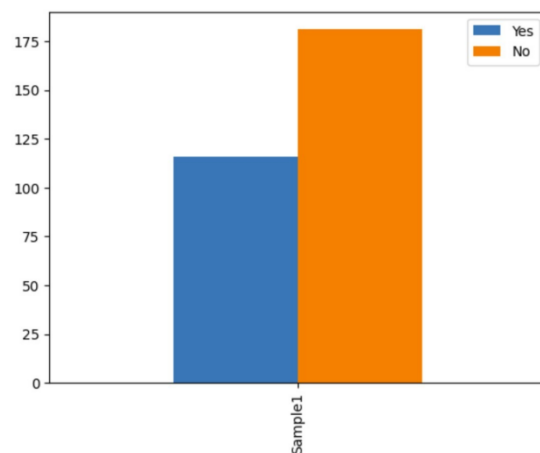


Figure: SUV purchases by personal loan status

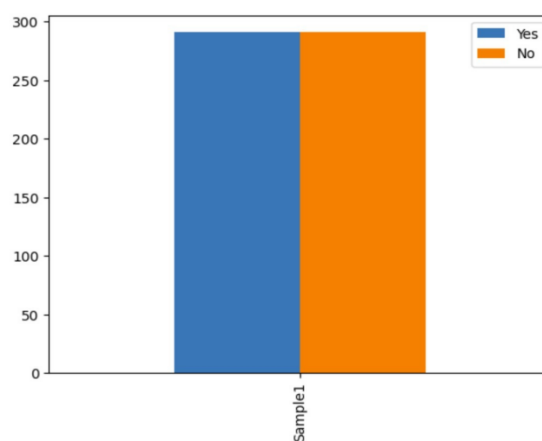


Figure: Hatchback purchases by personal loan status

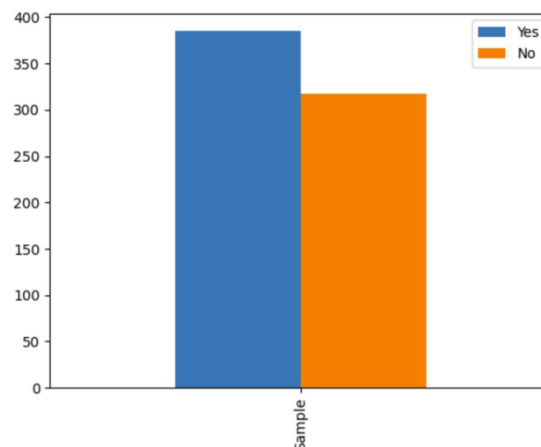


Figure: Sedan purchases by personal loan status

➤ House Loan vs Make

Analysis Performed

The dataset showed that customers **without a house loan were almost double** compared to those with a house loan. To address this imbalance, **two balanced samples** were created with equal counts of customers having and not having a house loan. The analysis was then carried out for **SUV, Hatchback, and Sedan** categories.

Observations

- **SUV:** Purchases by customers **without a house loan are significantly higher** than those with a house loan in both samples.
- **Hatchback:** Customers **with a house loan** show slightly higher purchases compared to those without a loan.
- **Sedan:** Sedan purchases are also **higher among customers with a house loan**, though the difference is not very large.

Overall Interpretation

House loan status clearly influences vehicle preference. Customers **without a house loan strongly prefer SUVs**, indicating higher disposable income or fewer financial commitments. In contrast, customers **with a house loan tend to prefer Hatchbacks and Sedans**, suggesting more conservative and budget-conscious purchasing behavior.

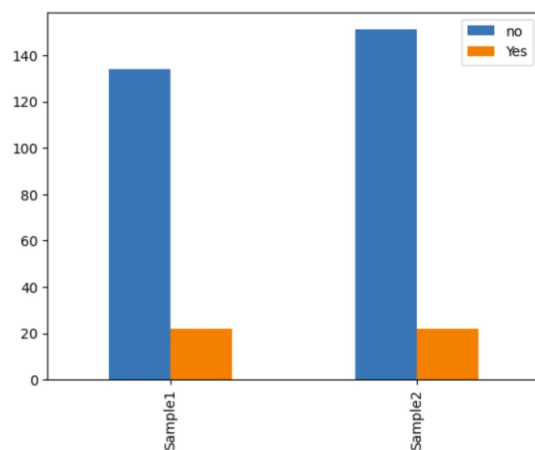


Figure: Sedan purchases by house loan status

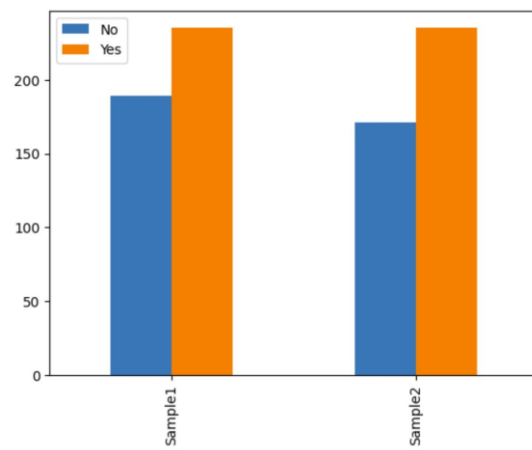


Figure: Sedan purchases by house loan status

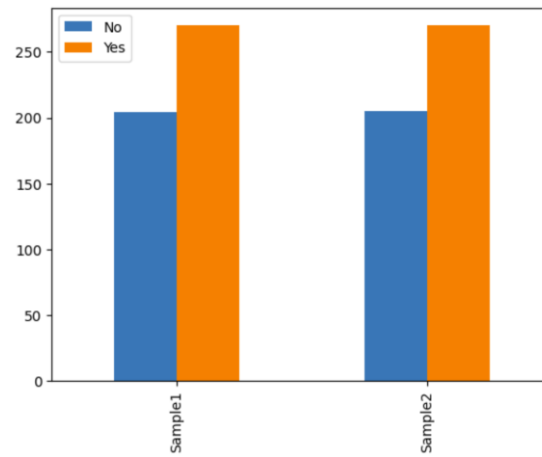


Figure: Sedan purchases by house loan status

Multivariate Analysis Approach: